Conjunction and derivability: projections and pairing, each one modus ponens from an axiom

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source cites ax:land1 in both halves of item (1)'s line - as printed, the second instance covers the B projection; kept verbatim. Two nested ollabels live inside the statement span.

Editorial note. The label prop:provability-land names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Conjunction elimination and introduction axioms. Propositional-logic part, axioms ax:land1 (twice, one per projection) and ax:land3 (pairing). Lines 1054 and 1056-1057: item (1) is 'modus ponens' on land1, item (2) 'two applications of modus ponens' on land3.

Source reference. Original reference ax-land1: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 494–494. Prerequisite pointer invoked at proof line(s) 1053; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-land3: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 496–496. Prerequisite pointer invoked at proof line(s) 1055; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *1. Both* $!A \wedge !B \vdash !A$ *and* $!A \wedge !B \vdash !B$

2. $!A, !B \vdash !A \wedge !B$.

Proof. 1. From ax-land1 and ax-land1 by modus ponens.

2. From ax-land3 by two applications of modus ponens.

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